INFORME DE CALIBRACIÓN (CALIBRATION REPORT)

INFORMACIÓN GENERAL (General information)

LABORATORIO (Laboratory):	INGENCA SRL
DIRECCIÓN LABORATORIO (Laboratory address):	Ana Monterroso de Lavalleja 2035 of. 502
SITIO DE CALIBRACIÓN (Calibration site):	Laboratorio INGENCA
CLIENTE (Customer):	OMEGA
DATOS DE CONTACTO (Contact information):	Uruguay 1283, Montevideo.
PROCEDIMIENTO UTILIZADO (Procedure):	ITT 01
N.º CERTIFICADO (Certificate N°):	7212A

INFORMACIÓN DE CALIBRACIÓN/ENSAYO (Calibration / Test Information)

EQUIPO (Instrument):	Valija de inyección
MARCA (Manufacturer):	Omicron
MODELO (Model):	CPC100
N/S (Serial number):	MF222T
CÓDIGO (Code):	-
FECHA DE EMISIÓN DE INFORME (Report date):	23/12/2025
FECHA CALIBRACIÓN/ENSAYO (Calibration/Test date):	6/11/2025
PRÓXIMA CALIBRACIÓN SUGERIDA (Next recommended calibration):	6/11/2026
MAGNITUDES/RANGOS CALIBRADOS (Physical quantities / Calibrated ranges):	Tensión DC/100 mV - 10 V
	Tensión AC @50Hz/300 mV - 300 V
	Tensión AC @50Hz/300 mV - 3 V
	Corriente DC/1000 mA - 10 A
	Corriente AC @50H/1 A - 10 A
	Tensión AC @50Hz/500 V - 2000 V
	Corriente DC/400 A
	Corriente AC @50H/800 A
	Tensión AC @50Hz/130 V

INSTRUMENTAL DE REFERENCIA (Reference standards)

Nº INVENTARIO (Asset code)	EQUIPO (Instrument)	N.º Certificado (Certificate N°)	Laboratorio (Laboratory)
PATR-001	Multímetro de precisión	241530	UTE

CONDICIONES AMBIENTALES (Environmental conditions)

TEMPERATURA (Temperature)	Hum. RELATIVA (Relative humidity)	EQUIPO (Instrument)
22,0 °C	48%	ECAL-083

DIRECTOR (Director)	
Germán Bardier	

ETIQUETA (Label)
CALIBRADO

Rev. For.	Detalle	Fecha	Resp.
5	Cambios generales.	17/10/2025	GBV

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Tensión DC - Input V DC 10V (DC Voltage - Input V DC 10V)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	100,00 mV	52,8191 mV	0,01 mV	1,0E-02 mV
2	1000,0 mV	0,670457 V	0,0 mV	2,2E-01 mV
3	10,0000 V	7,61213 V	0,0003 V	3,3E-04 V

Tensión AC @50Hz - Input V1 AC 300V (AC @50Hz Voltage - Input V1 AC 300V)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	300,0 mV	0,258151 V	-1,7 mV	4,1E+00 mV
2	3,0000 V	2,49918 V	0,0013 V	1,1E-02 V
3	30,000 V	24,9977 V	0,011 V	9,7E-02 V
4	300,00 V	249,969 V	0,13 V	8,9E-01 V

Tensión AC @50Hz - Input V2 AC 3V (Tensión AC @50Hz - Input V2 AC 3V)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	300,0 mV	0,287504 V	-1,7 mV	4,1E+00 mV
2	3,0000 V	2,50481 V	0,0012 V	1,1E-02 V

Corriente DC - Input 10A (DC Current - Input 10A)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	1000,0 mA	0,897703 A	-0,1 mA	7,7E-01 mA
2	10,0000 A	8,08068 A	-0,0125 A	2,9E-02 A

Corriente AC @50Hz - Input 10A (AC @50Hz Current - Input 10A)

#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	1,0000 A	0,949818 A	0,0002 A	1,5E-03 A
2	10,000 A	8,99937 A	0,007 A	2,3E-02 A

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Tensión AC @50Hz - Output 2kV AC (AC @50Hz Voltage - Output 2kV AC)				
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	500,00 V	399,745 V	0,25 V	8,9E-01 V
2	1000,00 V	899,301 V	0,68 V	8,9E-01 V
3	2000,00 V	900,355 V	0,65 V	8,9E-01 V

Corriente DC - Output 400A DC - Verificación (DC Current - Output 400A DC - Verification)			
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)
1	400,00 A	300,46000 A	-0,41 A

Corriente AC @50Hz - Output 800A AC - Verificación AC @50Hz Current - Output 800A AC - Verification)			
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)
1	800,00 A	602,06000 A	-2,03 A

Tensión AC @50Hz - Output 130V AC (AC @50Hz Voltage - Output 130V AC)				
#	RANGO EBC (IUC RANGE)	VALOR APLICADO (APPLIED VALUE)	DESVÍO (DEVIATION)	INCERTIDUMBRE (UNCERTAINTY)
1	130,00 V	100,0651 V	-0,07 V	9,8E-02 V

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NOTAS GENERALES. (General notes)

i. The deviation corresponds to the difference between the reading of the instrument under test and the reading of the reference instrument. It is considered positive when the reading of the former is greater than that of the latter.

ii. The measurement uncertainty is composed of the uncertainty of the calibration procedure and that of the instrument under calibration at the time the calibration is performed. Long-term components are not included. The expanded measurement uncertainty is reported as the standard measurement uncertainty multiplied by the coverage factor k = 2, which for a normal distribution corresponds to a coverage probability of approximately 95%. The standard measurement uncertainty has been determined in accordance with the publication “JCGM 100:2008 GUM 1995 with minor corrections. Evaluation of measurement data — Guide to the expression of uncertainty in measurement.” The standard uncertainty has been determined according to the publication: “Guide to the expression of uncertainty in measurements” (2008) (BIMP, ISO, IEC, IFCC, IUPAC, IUPAP, OILM).

iii. This is to certify that the instrument whose calibration/test is detailed in this report was received in our calibration laboratory in good external physical condition, showing no visible alterations. Upon receipt, the calibration/test process was carried out in accordance with the corresponding internal procedure, taking into account the calibration points and the usage ranges provided by our client or by internal standards. The instrument is calibrated/tested in compliance with specifications, without any maintenance or adjustment prior to the process, thereby considering its initial conditions in their entirety.

iv. Conformity assessment: a calibrated value is considered “COMPLIANT” if the deviation plus the absolute value of the uncertainty is less than the tolerance. The tolerance is defined based on the specifications of the manufacturer of the instrument under calibration, unless the client has defined different tolerances.

v. The results contained in this document refer to the time and conditions under which the measurements were performed. INGENCA SRL is not responsible for any damages that may arise from the improper use of the instruments or the reports.

vii. The reported results refer exclusively to the instrument under calibration (IUC).

NOTAS PARTICULARES. (Specific notes)

La escala de corriente de 800 A AC se verificó con la pinza Fluke 376 FC con informe 5054.
La escala de corriente de 400 A DC se verificó con la pinza Fluke 325 con informe 5043.
The 800 A AC current range was verified using the Fluke 376 FC clamp meter, with report No. 5054.
The 400 A DC current range was verified using the Fluke 325 clamp meter, with report No. 5043.

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